

**LAB NUMBER & TITLE: LAB 01 - UBUNTU SERVER
INSTALLATION AND SSH CONFIGURATION**

NAME: SHANZA ANAM

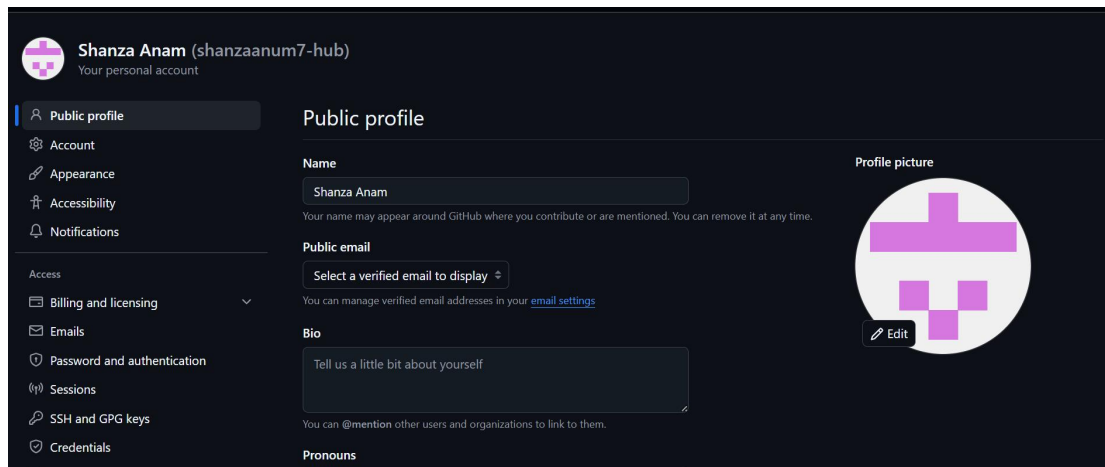
REGISTRATION NUMBER: FA24B1-SE-049

SECTION: 5A

COURSE CODE: CLOUD COMPUTING(SE-665)

GITHUB USERNAME: SHANZAANUM7-HUB

Task 1: Create and Configure Your GitHub Account

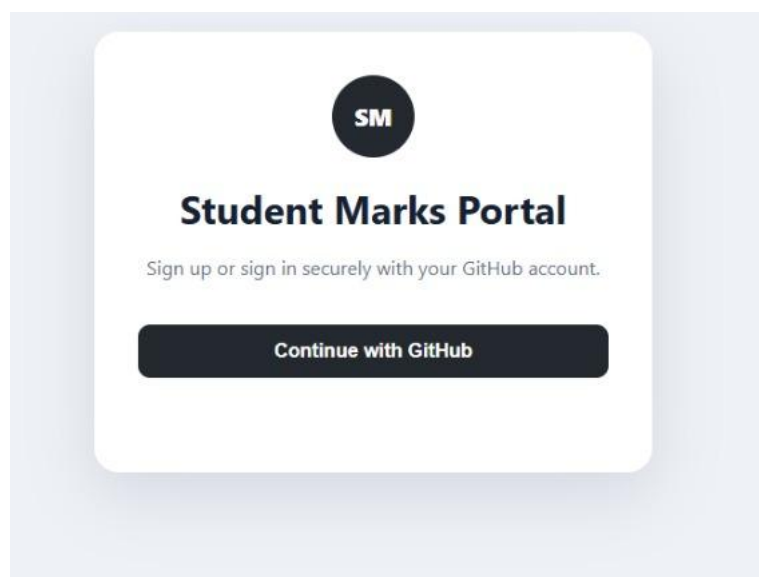


GitHub profile showing my name and username.

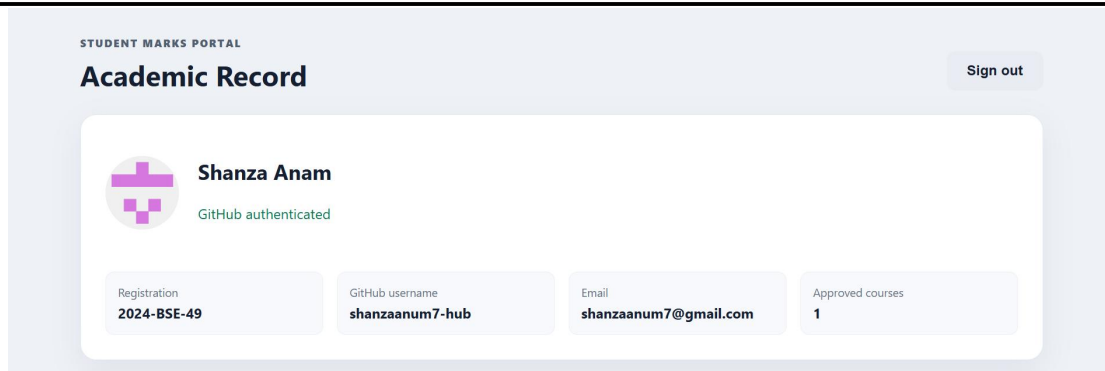


My actual name is entered in the GitHub public profile.

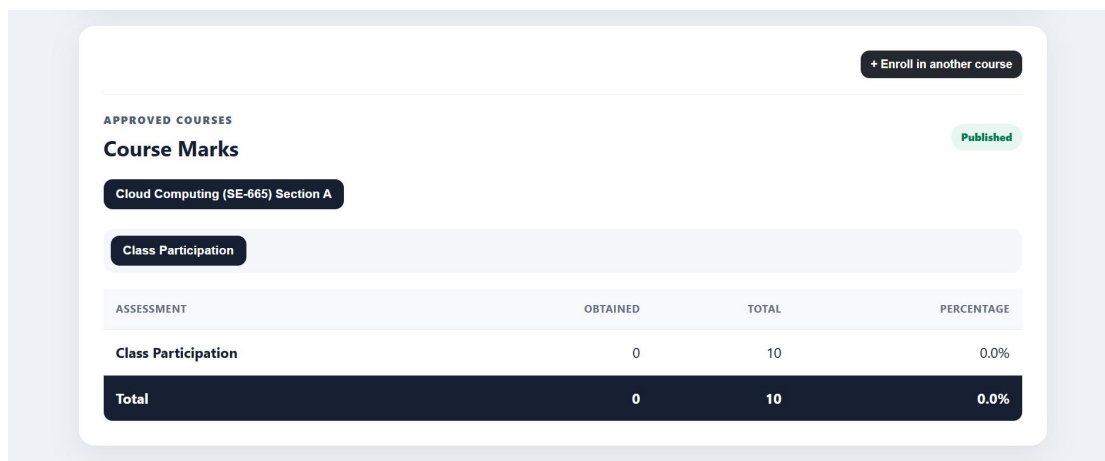
Task 2: Register in the Student Marks Portal



Student Marks Portal login page before signing in.

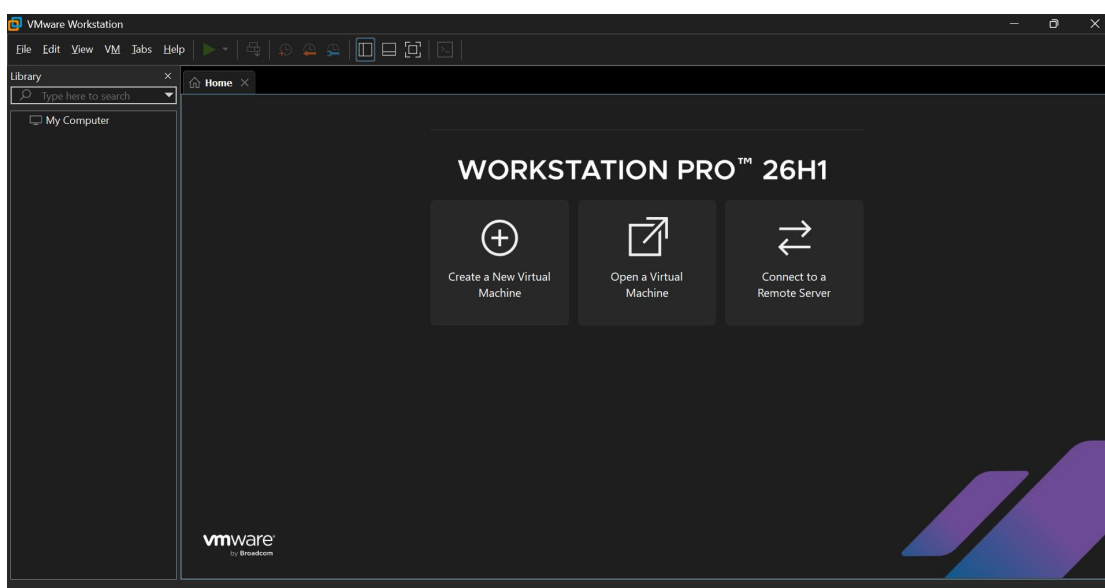


My GitHub profile information is shown in the student portal.

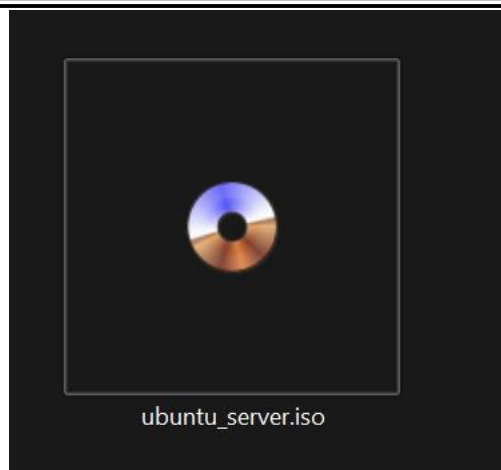


The course enrollment has been submitted for approval.

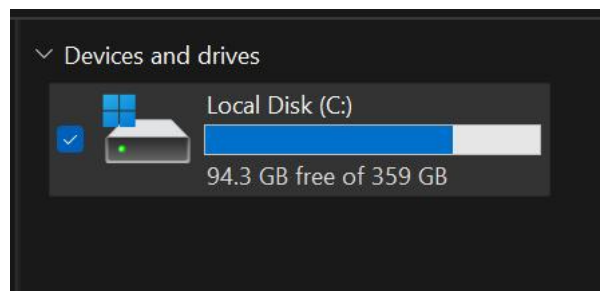
Task 3: Download VMware Workstation Pro and Ubuntu Server



VMware Workstation Pro is installed and running on my computer.



The downloaded Ubuntu Server LTS ISO file is shown here.



The available storage is checked before creating the virtual machine.

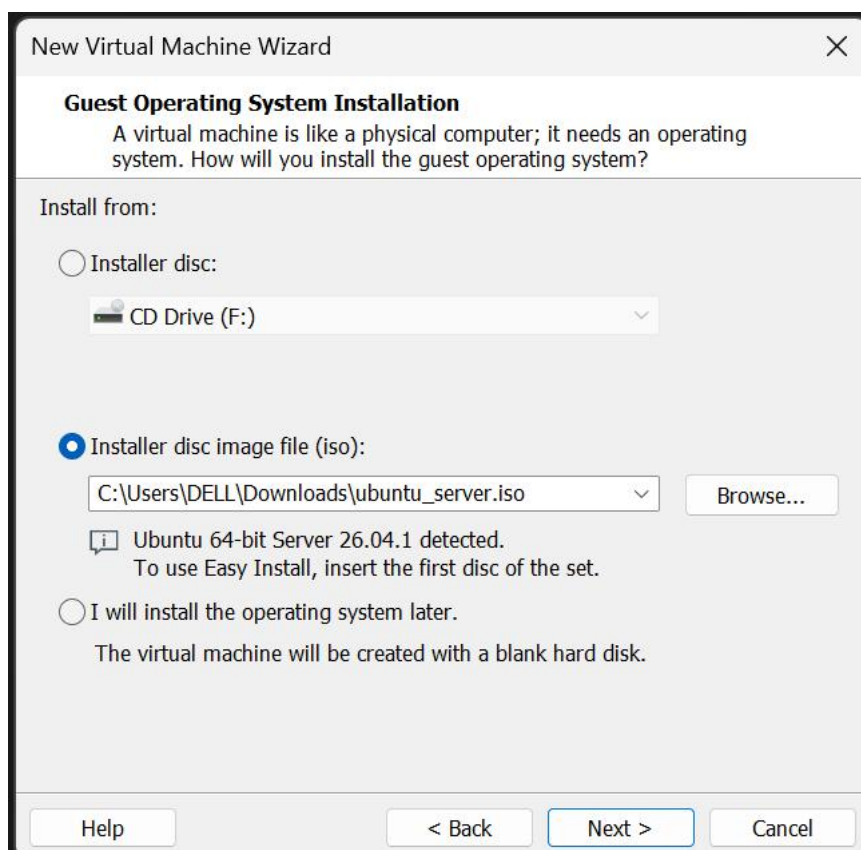
Task 4: Create the Ubuntu Server Virtual Machine



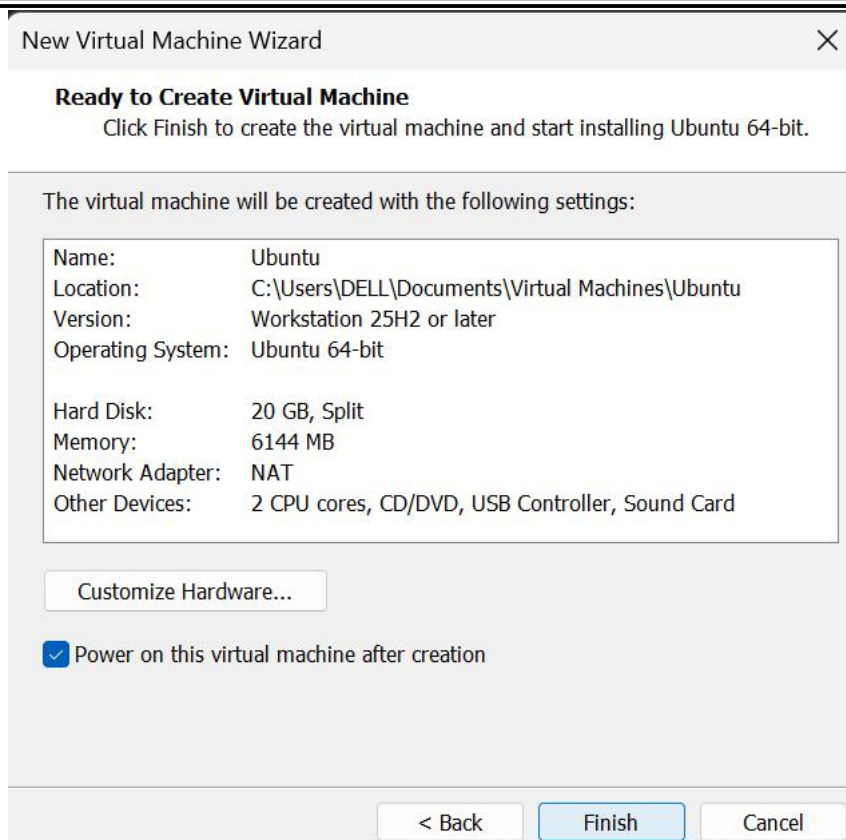
The VMware New Virtual Machine Wizard is open.



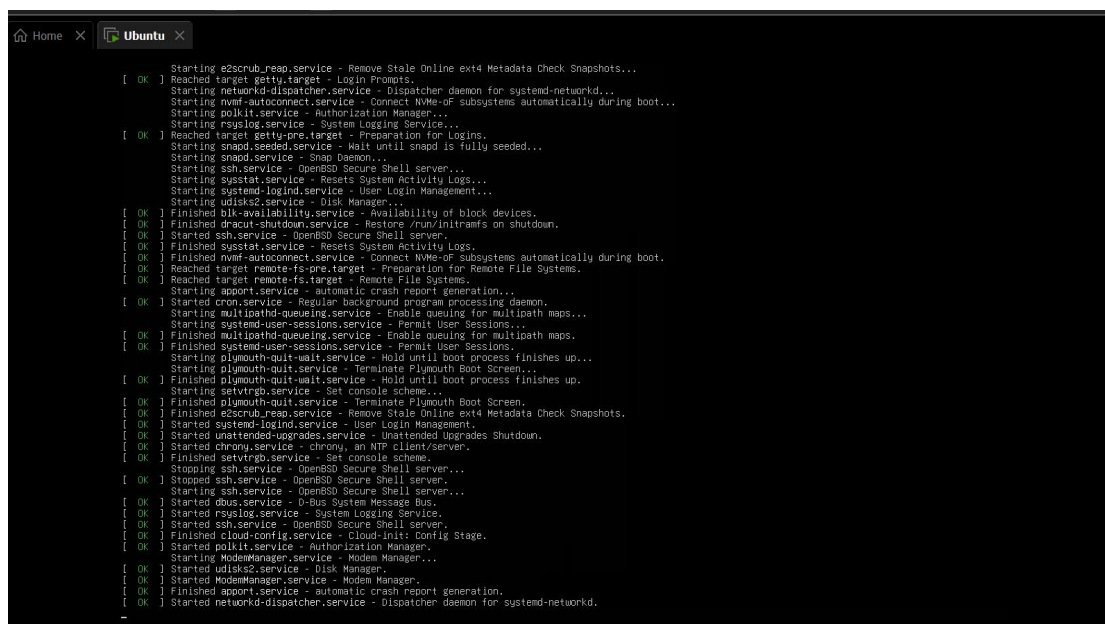
The Typical configuration option is selected for the virtual machine.



The Ubuntu Server ISO is selected as the installation media.

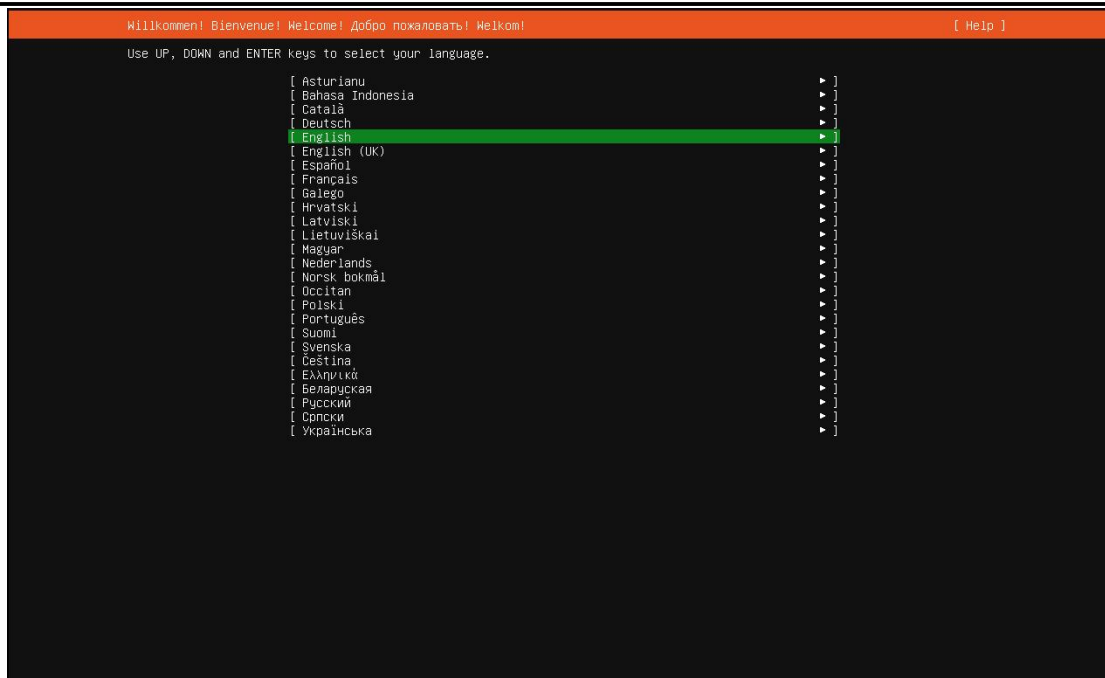


The final virtual machine settings are shown before completing the setup.



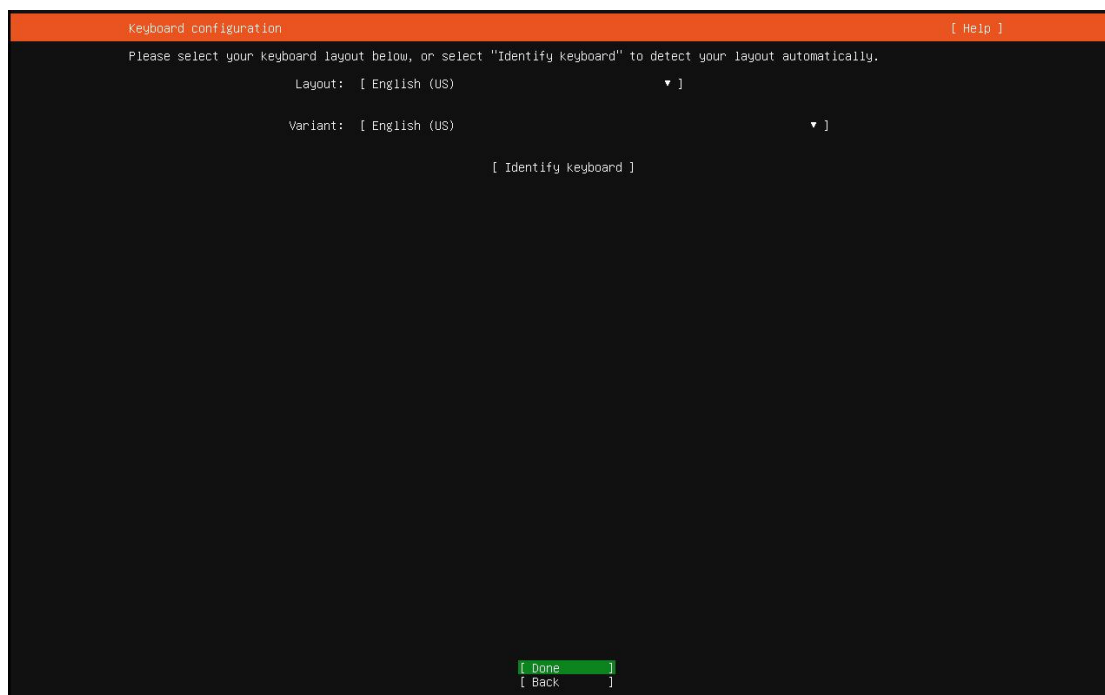
The Ubuntu Server installer has started after powering on the virtual machine.

Task 5: Choose the Installation Language

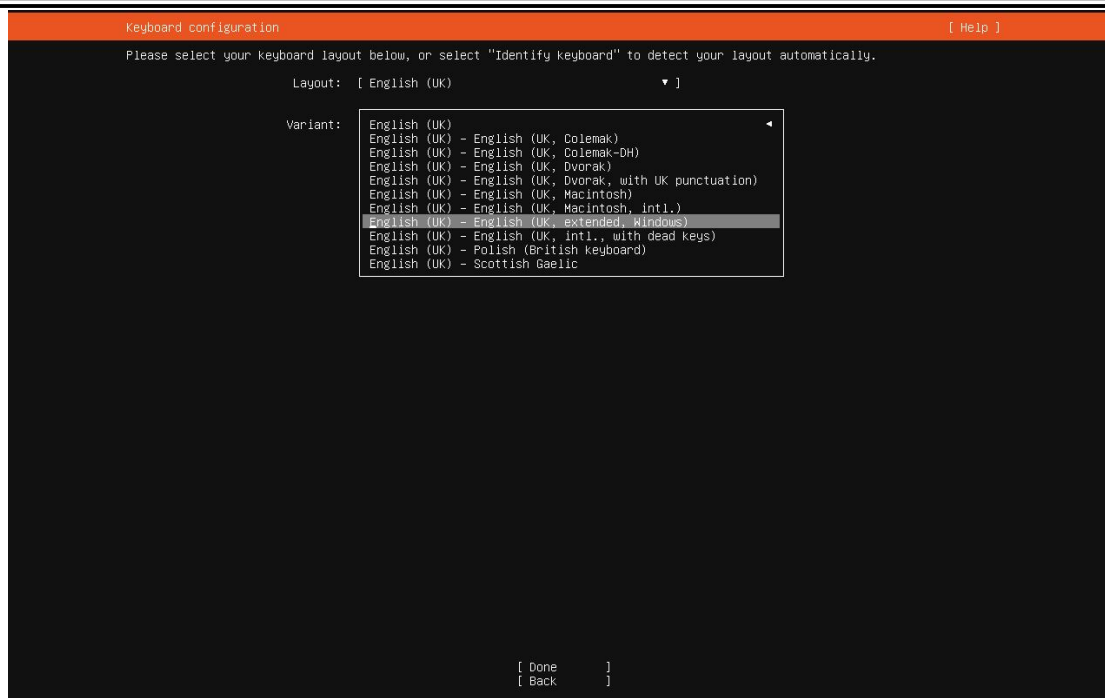


The installation language is selected for Ubuntu Server.

Task 6: Configure the Keyboard Layout

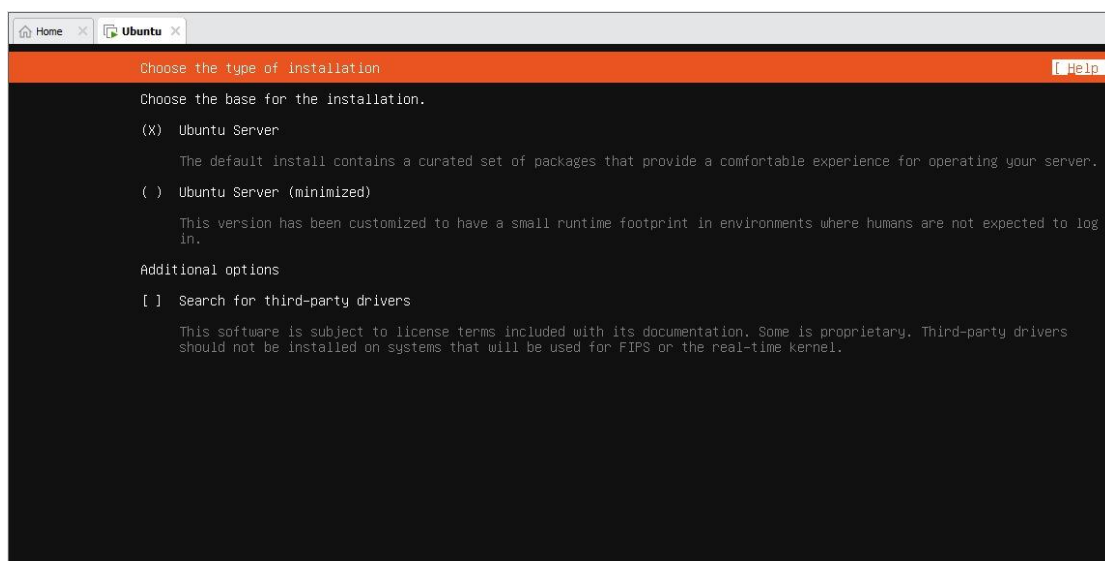


The keyboard layout is selected according to my keyboard.



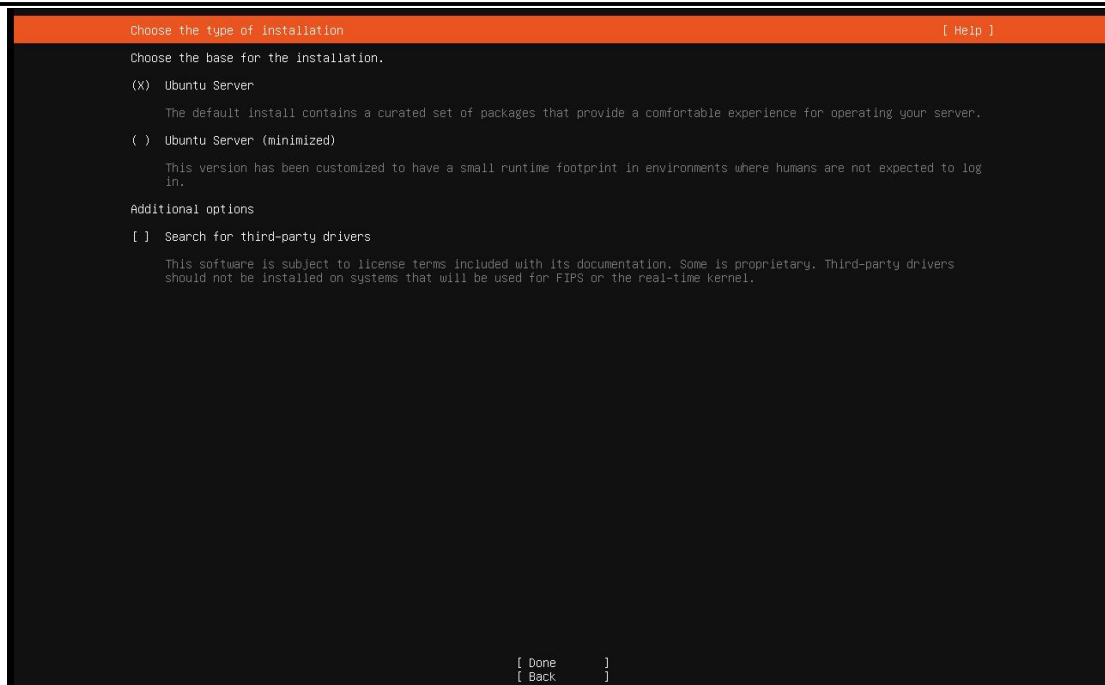
The appropriate keyboard variant is selected.

Task 7: Choose the Ubuntu Server Installation Type

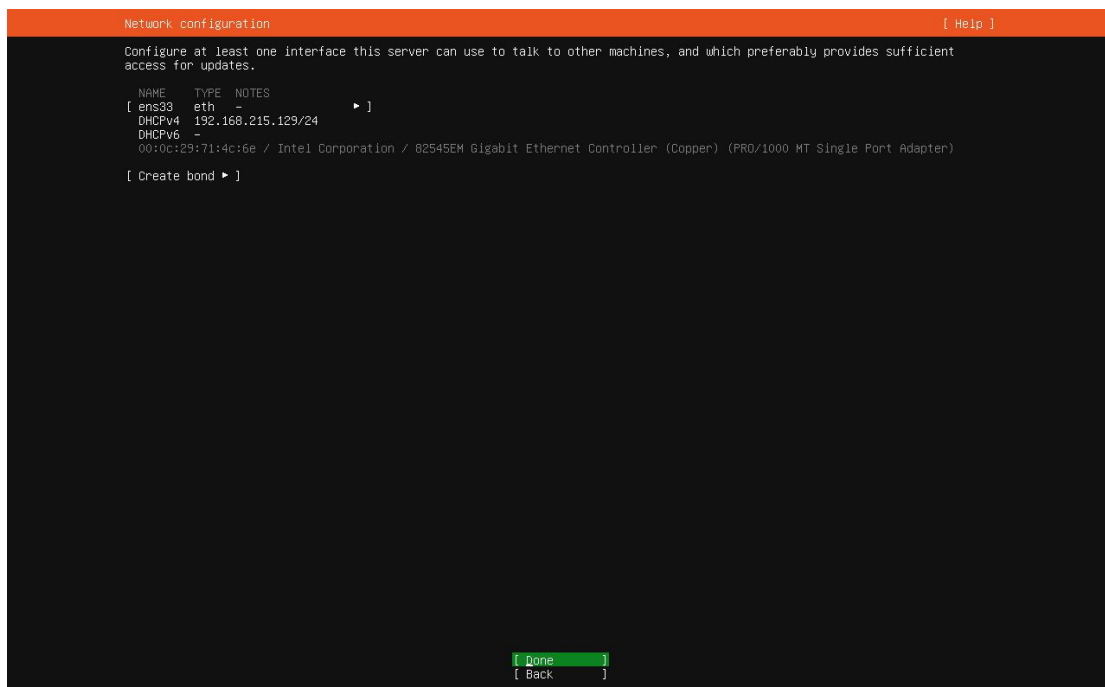


The normal Ubuntu Server installation option is selected.

Task 8: Configure the Network



The network interface detected by the Ubuntu Server installer is shown.



The Ubuntu Server installer has received an IP address through DHCP.

Task 9: Configure Storage

Proxy configuration

[Help]

If this system requires a proxy to connect to the internet, enter its details here.

Proxy address:

If you need to use a HTTP proxy to access the outside world, enter the proxy information here. Otherwise, leave this blank.

The proxy information should be given in the standard form of "http://[user][:pass]@host[:port]/".

[Done]

[Back]

Guided storage is selected for the VMware virtual disk.

Guided storage configuration

[Help]

Configure a guided storage layout, or create a custom one:

(X) Use an entire disk

[/dev/sda local disk 20.000G ▼]

[X] Set up this disk as an LVM group

[] Encrypt the LVM group with LUKS

Passphrase:

Confirm passphrase:

[] Also create a recovery key
The key will be stored as ~/recovery-key.txt in the live system and will be copied to /var/log/installer/ in the target system.

() Custom storage layout

[Done]

[Back]

The proposed storage configuration is reviewed before continuing.

Task 10: Select the Installation Disk

USED DEVICES

```
DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                    LVM volume group                  18.222G ▶ ]
ubuntu-lv    new, to be formatted as ext4, mounted at /  10.000G ▶ ]

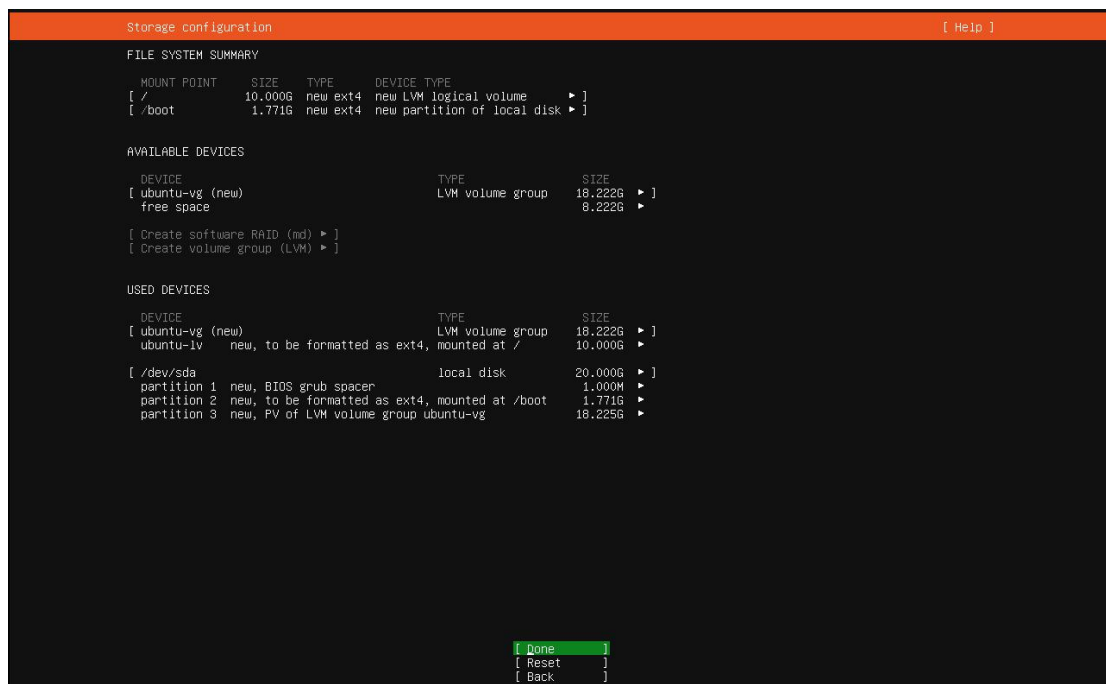
[ /dev/sda                            local disk                        20.000G ▶ ]
partition 1  new, BIOS grub spacer    1.000M ▶ ]
partition 2  new, to be formatted as ext4, mounted at /boot  1.771G ▶ ]
partition 3  new, PV of LVM volume group ubuntu-vg          18.225G ▶ ]
```

The VMware virtual disk is selected for the Ubuntu Server installation.

```
ubuntu-lv    new, to be formatted as ext4, mounted at /  10.000G ▶ ]
[ /dev/sda                            local disk                        20.000G ▶ ]
partition 1  new, BIOS grub spacer    1.000M ▶ ]
partition 2  new, to be formatted as ext4, mounted at /boot  1.771G ▶ ]
partition 3  new, PV of LVM volume group ubuntu-vg          18.225G ▶ ]
```

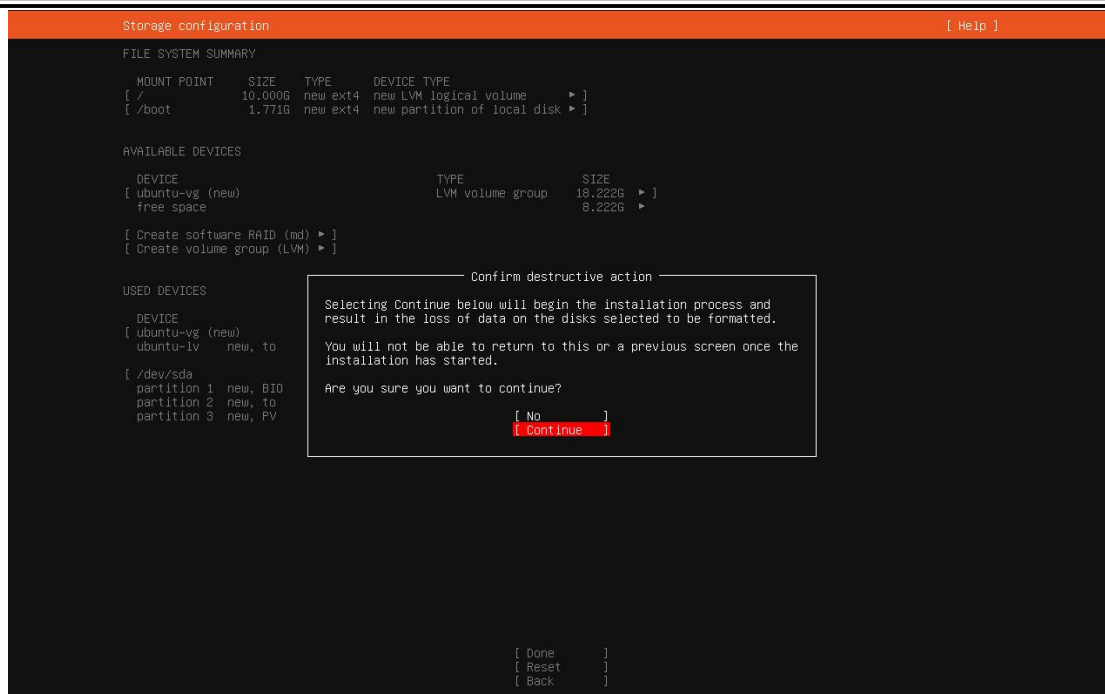
The selected virtual disk and its size are shown here.

Task 11: Review the Partitions

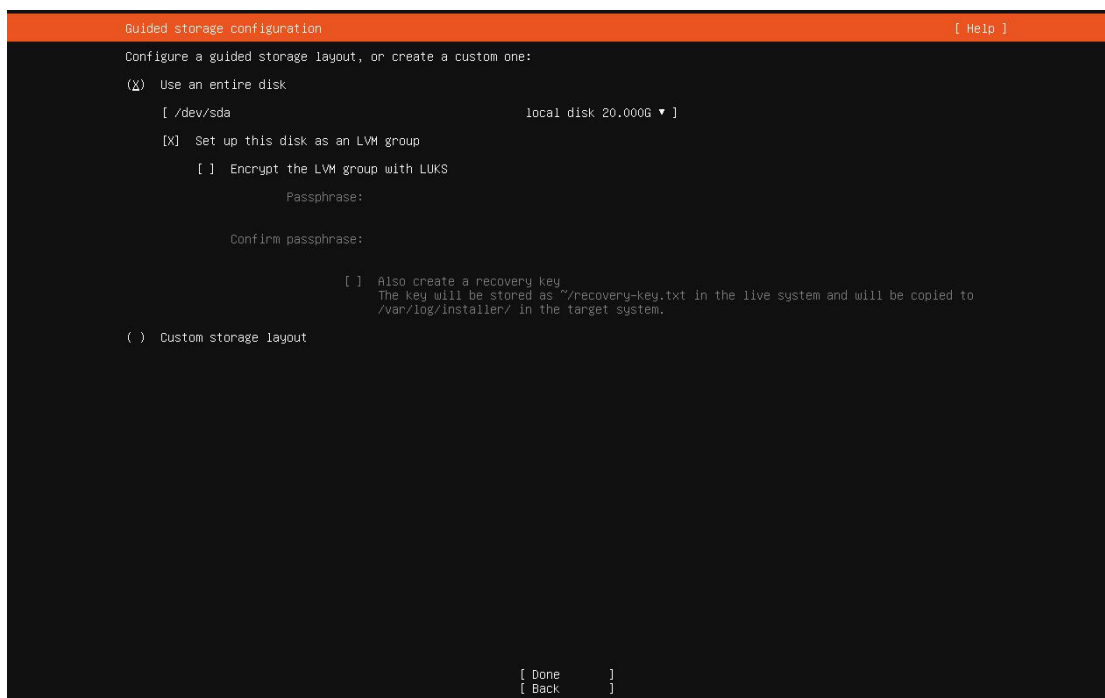


The partition layout proposed for the virtual disk is shown.

Task 12: Confirm the Storage Changes



The installer displays the warning before applying the storage changes.



The storage changes are confirmed for the VMware virtual disk.

Task 13: Configure the Ubuntu Profile

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:

Server name must not be empty

Pick a username:

Choose a password:

Confirm your password:

[Done]

My complete name is entered in the Ubuntu profile.

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:

The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

The server name is set to ubuntu as required.

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:

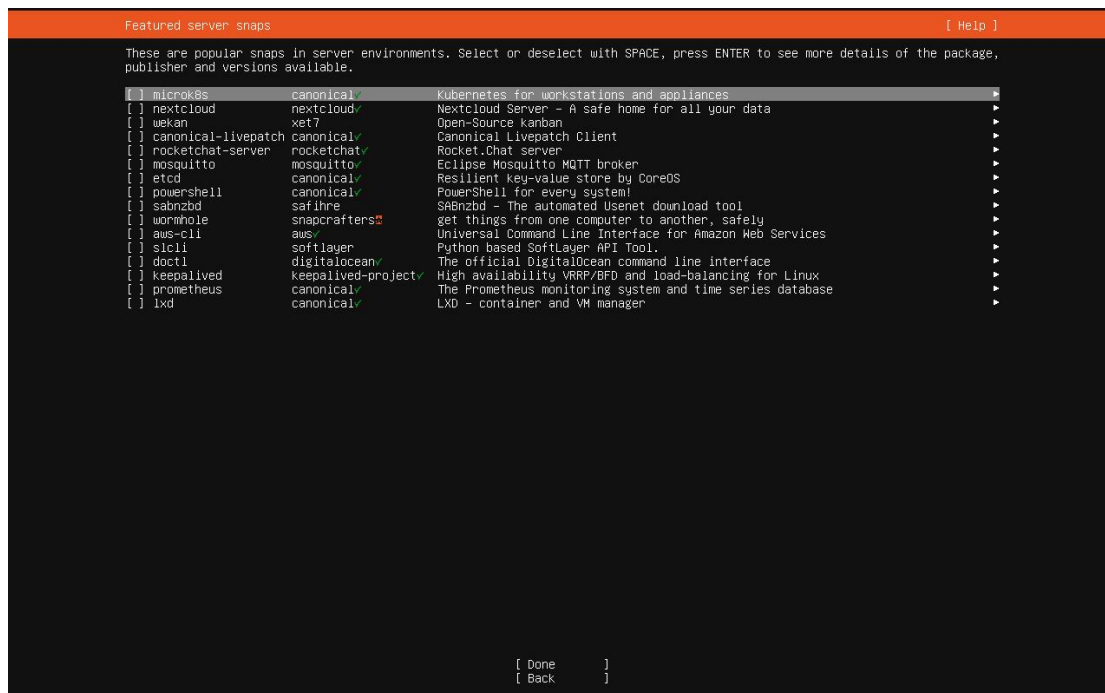
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

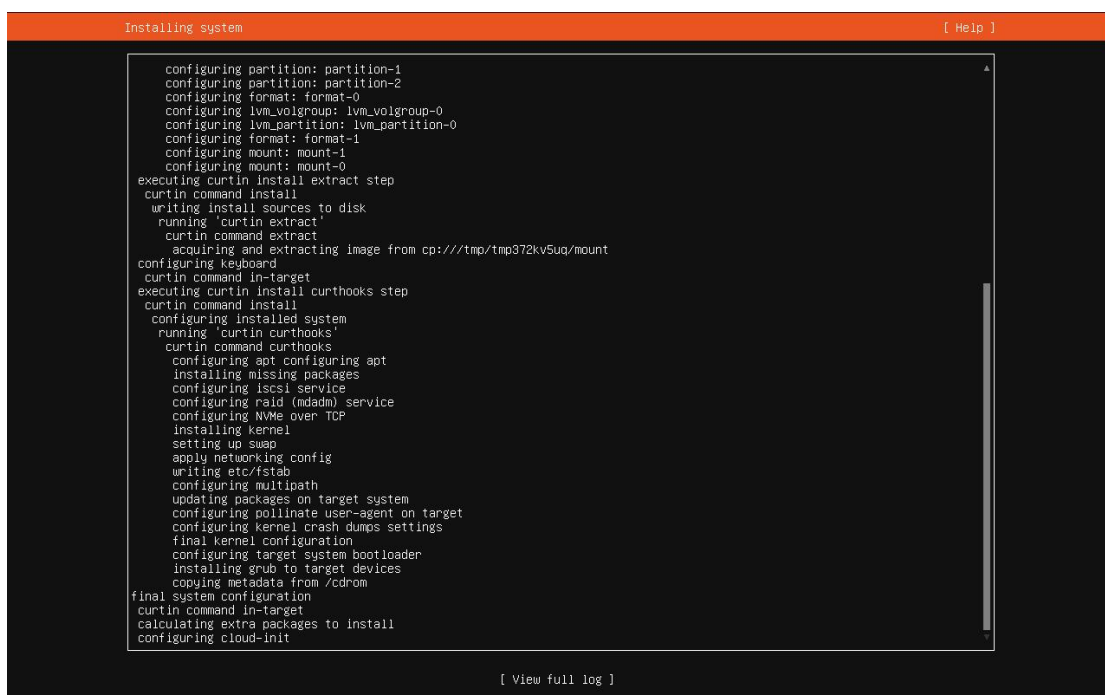
Confirm your password:

My GitHub username is entered as the Ubuntu username.



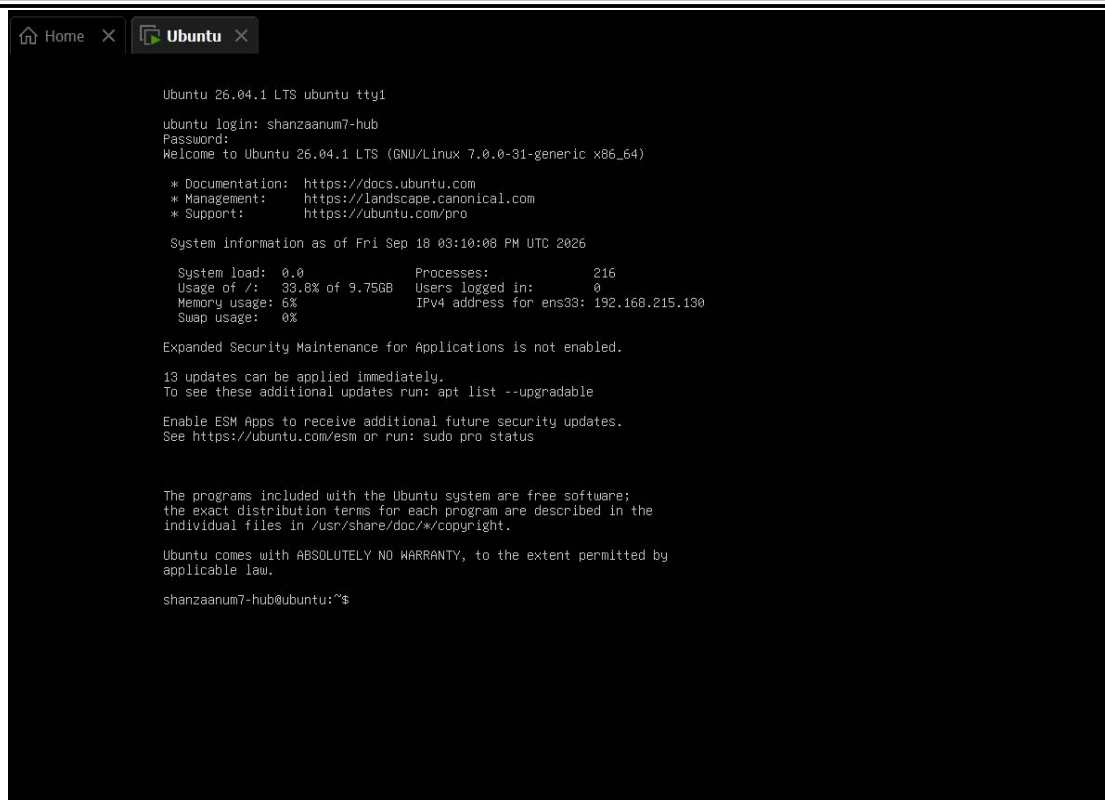
The Ubuntu profile is completed with the required information.

Task 14: Complete the Software Installation

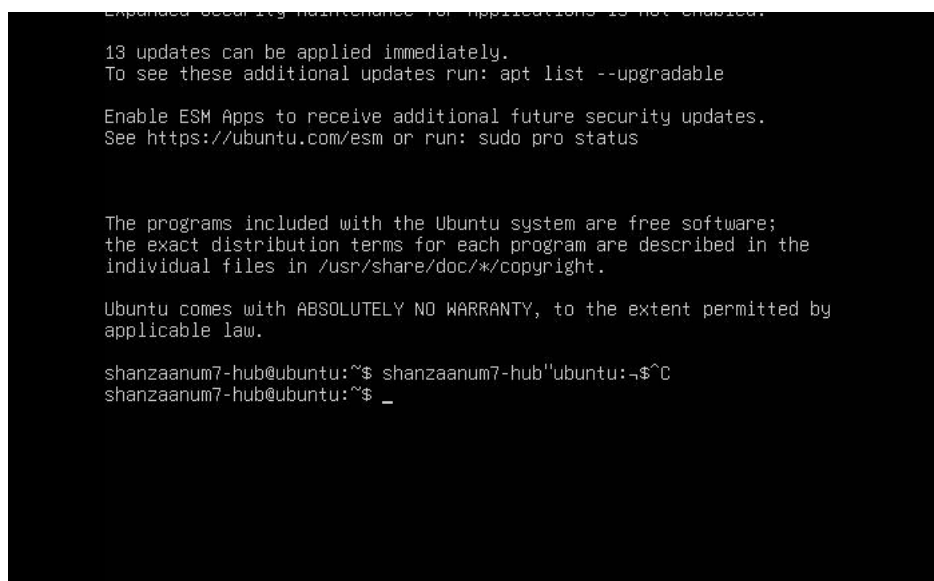


Ubuntu Server is installing the required software.

Task 15: Reboot and Verify the Ubuntu Identity

A screenshot of a terminal window titled 'Ubuntu'. The terminal shows the login process for 'shanzaanum7-hub' on 'Ubuntu 26.04.1 LTS'. It displays system information such as system load, memory usage, and the number of processes. It also mentions that expanded security maintenance for applications is not enabled and lists 13 updates that can be applied immediately. The terminal ends with the prompt 'shanzaanum7-hub@ubuntu:~\$'.

I have successfully logged in to the Ubuntu Server terminal.

A screenshot of the terminal window showing the output of the 'apt list --upgradable' command, which indicates that 13 updates can be applied immediately. It also shows the command 'sudo pro status' and its output, which states that the programs included with the Ubuntu system are free software and that Ubuntu comes with ABSOLUTELY NO WARRANTY. The terminal ends with the prompt 'shanzaanum7-hub@ubuntu:~\$'.

The terminal prompt confirms my Ubuntu username and server name.

Task 16: Find the Ubuntu Server IP Address

```

shanzaanum7-hub@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:71:4c:6e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29714c6e
    inet 192.168.215.130/24 metric 100 brd 192.168.215.255 scope global dynamic ens33
        valid_lft 991sec preferred_lft 991sec
    inet6 fe80::29c:29ff:fe71:4c6e/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
shanzaanum7-hub@ubuntu:~$

```

The ip addr command is used to check the server network information.

```

2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:71:4c:6e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29714c6e
    inet 192.168.215.130/24 metric 100 brd 192.168.215.255 scope global dynamic ens33
        valid_lft 991sec preferred_lft 991sec
    inet6 fe80::29c:29ff:fe71:4c6e/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
shanzaanum7-hub@ubuntu:~$ _

```

The active network interface shows the IPv4 address of the Ubuntu Server.

Task 17: Connect to Ubuntu Server from Windows

```

C:\Users\DELL>ssh shanzaanum7-hub@192.168.215.130
The authenticity of host '192.168.215.130 (192.168.215.130)' can't be established.
ED25519 key fingerprint is SHA256:4JhU1ji+dWB/Fb21k8rBFbVnBBp+sznQmkgF7BozbHE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.215.130' (ED25519) to the list of known hosts.
shanzaanum7-hub@192.168.215.130's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Sep 18 03:44:31 PM UTC 2026

System load:  0.02               Processes:            224
Usage of /:   34.1% of 9.75GB    Users logged in:     0
Memory usage: 7%                IPv4 address for ens33: 192.168.215.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.

```

The SSH command is entered in Windows to connect to the Ubuntu Server.

```

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

shanzaanum7-hub@ubuntu:~$ |

```

The server fingerprint is accepted to continue the SSH connection.

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
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Expanded Security Maintenance for Applications is not enabled.

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To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

shanzaanum7-hub@ubuntu:~$ |
```

The Windows terminal is successfully connected to the Ubuntu Server.

```
See https://ubuntu.com/esm or run: sudo pro status

shanzaanum7-hub@ubuntu:~$ |
```

The remote terminal prompt confirms the Ubuntu server identity.